Fiscal Year:	FY 2007	**Task Last Updated:**	FY 04/10/2007
PI Name:	Shaw, Benjamin Ph.D.		
Project Title:	Combustion Experiments in Reduced Gravity with Two-Component Miscible Droplets		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	95616-5294	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	08/30/2004	**End Date:**	12/21/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04AA65A		
Performance Goal No.:	1.22		
Performance Goal Text:	Enable increased combustion system efficiency, reduced pollution, and mitigation of fire risks through insights and data bases obtainable only through microgravity experiments.		
Task Description:	The original objective of this research was to study the combustion of bi-component miscible droplets in reduced-gravity environments so that buoyancy effects are rendered negligible. The program was termed the Bi-Component Droplet Combustion Experiment (BCDCE). Of particular interest were the burning rate (droplet history), the time to flame contraction, potential microexplosions, Marangoni flows, and sooting behaviors. The experiments used liquid mixtures selected to display influences of large component volatility differences and liquid species diffusion on combustion phenomena. The program was both experimental and theoretical in nature. Experiments included reduced-gravity studies as well as normal-gravity comparison experiments. In fall 2005, the BCDCE program was aligned with a different program that includes a team of investigators from different universities as well as from NASA. This program, which is termed the Flammability and Extinction		

	Experiment (FLEX), is focused upon using droplets to study flammability and extinction under reduced-gravity conditions of relevance to spaceflight and human exploration of space. The FLEX program is both experimental and theoretical in nature. Experiments include reduced-gravity studies as well as normal-gravity comparison experiments. In the research, individual droplets are dispensed, deployed and ignited in reduced gravity (as well as normal gravity). Theoretical approaches include developing computational and analytical models, with a particular emphasis on evaluating influences of gas-phase diffusive transport as well as droplet internal flows, which may arise from Marangoni stresses as well as droplet deployment and ignition.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The BCDCE research extended previous research on combustion of pure liquid fuels to mixtures of pure fuels, which more closely approximate practical fuels. The results from this research program were relevant to design of practical combustion systems that employ liquid fuels such as engines and oil-fired burners. The FLEX results will be useful for improving understanding of combustion and fire-related phenomena under reduced gravity conditions, e.g., influences of diffusive transport and droplet internal flows on flammability and extinction.
Task Progress:	Reduced gravity and normal gravity combustion experiments were performed with individual droplets that were initially about 1 mm in diameter. Most experiments were performed with propanol or heptane, but some data were also obtained with methanol. The environment was composed of an air-inert mixture at 0.1 MPa and about 298 K, where helium, carbon dioxide and xenon were separately used as inerts. The amount of inert was progressively increased until droplets could no longer be ignited. The amount of inert gas required to suppress combustion was generally higher in reduced gravity than normal gravity. The data show that ambient gas transport properties such as Soret transport play an important role in determining flammability of droplets. The data also showed that flammability characteristics depend on the gravitational level and fuel type. Analysis of previous reduced-gravity experiments involved digitizing and analyzing video images from experiments on combustion of individual propanol droplets. These experiments were conducted in air at standard temperature and with pressures ranging from 0.1 MPa to 1.0 MPa. The data show that sooting characteristics, burning rates and flame shapes vary significantly with pressure. Calculations indicate that burning rates are controlled by the liquid temperature, which increases with increasing pressure. Computational modeling involved extending a previous 1-dimensional numerical model to account for non-unity activity coefficients for propanol-water and propanol-water-glycerol mixtures. Calculations indicate that water absorption is very small for droplets burning in air at pressures ranging from 0.1 MPa to 1.0 MPa. A multidimensional computational model was also further developed. The code is based on overset grid technology to allow modeling of a droplet on a fiber. The code was used for droplet evaporation or combustion studies, where capillary forces were accounted for in some calculations. Results demonstrate that droplet vaporization characteristics depend on the fiber diameter and that Marangoni stresses can also be important. Three-dimensional calculations of fiber supported, or unsupported, 100 micron methanol droplets burning at elevated temperature and pressure showed that flame shapes and the onset time for ignition depend on whether a support fiber is present. Analytical modeling focused on predicting influences of fuel pyrolysis on droplet combustion characteristics. Transfer numbers, burning rates and quasisteady flame standoff ratios can be appreciably reduced when fuel pyrolysis effects are included in the analyses. It was also found that fuel pyrolysis can influence temperature gradients between droplets and flames, which influences formation of soot shells. The NASA Tracker Station was used to capture and analyze 16 mm films from droplet combustion experiments. A high resolution digital camera and a macro lens were used to form an automated system where individual frames of film were converted into high-resolution digital images. Custom computer codes were written for analysis of the images. Parameters of interest such as droplet and flame diameters can be efficiently determined with high resolution.
Bibliography Type:	Description: (Last Updated: 04/29/2010)
Articles in Peer-reviewed Journals	Shaw BD, Wei JB. "Propanol droplet flammability and combustion in air-diluent environments under normal and reduced gravity." Combustion Science and Technology, 2007 Jun;179(6):1205-23. http://dx.doi.org/10.1080/00102200601057394 , Jun-2007
Articles in Peer-reviewed Journals	Dakka SM, Shaw BD. "Influences of pressure on reduced-gravity combustion of 1-propanol droplet." Microgravity Science and Technology 2006;18(2): 5-13. , Jun-2006
Papers from Meeting Proceedings	Shaw BD, Wei JB. "Influence of Gravity on Combustion and Flammability of Heptane Droplets in Air-Diluent Environments." 5th US Combustion Meeting, University of California at San Diego, La Jolla, CA, March 25-28, 2007. 5th US Combustion Meeting, March 2007, Paper G17. , Mar-2007
Papers from Meeting Proceedings	Xiao JS, Wei JB, Shaw BD, Peng Z. "Composition and Temperature Measurements of Evaporating Propanol-Acetone Droplets." 5th US Combustion Meeting, University of California at San Diego, La Jolla, CA, March 25-28, 2007. 5th US Combustion Meeting, March 2007, Paper G13. , Mar-2007